Web Fundamentals

Program Code: XE107

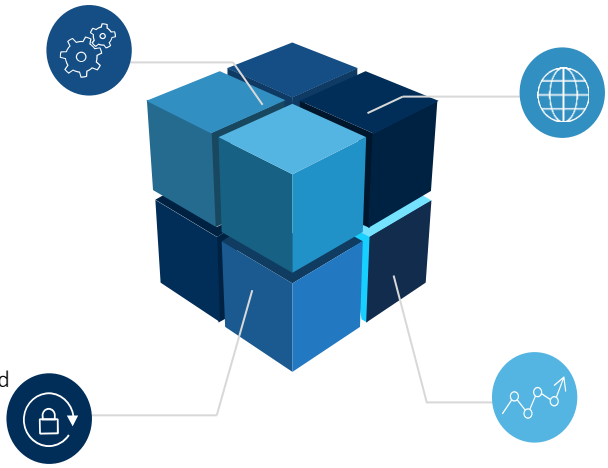
OPERATION ORDER

Operation **Phishing Framework** involves creating an automated Bash-based system designed for educational purposes, enabling cybersecurity students to simulate phishing attacks practically. The framework automates downloading target web pages, embedding credential collection mechanisms, redirecting users to legitimate sites, configuring SSL certificates conditionally based on domain availability, and managing Apache servers and shortened URLs. The goal is demonstrating realistic phishing scenarios securely and responsibly.



1. VISION

Equip cybersecurity students with practical experience in recognizing, understanding, and simulating phishing techniques within controlled educational environments, improving their ability to detect and defend against such threats.



2. MISSION

Develop a comprehensive Bash-driven phishing simulation framework capable of automating webpage capture and modification, credential harvesting, conditional SSL deployment, server management, and realistic phishing URL generation.

3. STRATEGY

- Automate webpage retrieval and duplication using wget.
- Automatically modify webpage forms to integrate PHP scripts for capturing credentials.
- Redirect users immediately to the legitimate target website after credential submission.
- Conditionally automate SSL certificate configuration depending on user's domain availability.
- Automate Apache server deployment and management.

4. OBJECTIVES

- Auto-download and modify webpages with PHP stealers.
- Log creds with IP and timestamp.
- Redirect users post-capture.
- Auto SSL if domain given, else use HTTP.
- Auto-configure Apache.
- Auto-generate short phishing links.

Project Structure

1. Dependency Check and Installation

1.1 Check for required tools
1.2 Install missing tools automatically
1.3 Skip installation for tools already installed

2. Choose Target and Setup Template

2.1 Ask user to choose one option:

- 2.1.1 Clone a website (enter URL)
- 2.1.2 Built-in template (Facebook, Gmail)

2.2 If cloning a website:

- 2.2.1 Download webpage with wget
- 2.2.2 Inject PHP script automatically for capturing credentials

2.3 If using a built-in template:

- 2.3.1 Automatically copy template files (HTML + PHP) to working directory

3. Domain and SSL Setup

3.1 Prompt user: "Do you have a custom domain?"
3.2 If "Yes":

- 3.2.1 Ask user to enter domain name
- 3.2.2 Automatically configure SSL certificate

3.3 If "No":

- 3.3.1 Configure phishing page over HTTP (port 80)

4. Credential Capture and User Redirection

4.1 Use PHP script to capture submitted credentials (include timestamp and IP)
4.2 Automatically redirect user to original website after submission
4.3 Save credentials locally in CSV or JSON format

5. Apache Server Configuration

5.1 Automatically configure Apache server to host phishing pages
5.2 Adjust Apache settings based on SSL usage
5.3 Restart Apache automatically to apply changes

6. Short URL Creation

6.1 Automatically create public URL for phishing page
6.2 Display shortened URL after script completion

General

- **Tools Required:** wget, Apache2, Certbot, PHP, Bitly or TinyURL APIs
- Automate all tasks fully, except minimal user input
- Use modular, clearly-defined functions for readability and maintainability

Comments and Documentation

Use comments in your code to explain what you did. If you are using code from the internet, add credit and links. In the script, write the student's name and code, the class code, and the lecturer's name.

Submitting Your Project

To ensure a successful submission of your project, please adhere to the instructions provided below. Following these guidelines will facilitate a smooth evaluation process and allow you to receive feedback on your work promptly.

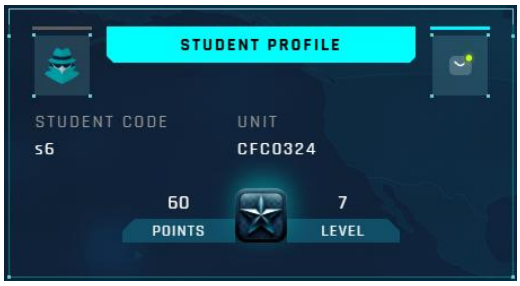
1. What to Submit

Only .sh and .pdf files will be accepted. **No other file types** (e.g., .docx, .txt) will be accepted. Ensure your files are in the correct format before submission.

You must submit both the script (.sh) and the corresponding PDF. The PDF should prove that you completed all the tasks and that the project works as expected. If you miss the script file or the PDF, you will be asked to resend your submission.

2. File Naming

Ensure your project files follow the correct naming convention:
UNIT.STUDENT.PROGRAM



For example, if your unit is CFC0324, your student ID is s6, and your program code is nx201, the correct filenames should be:

CFC0324.s6.xe107.sh
CFC0324.s6.xe107.pdf

3. Submission Process

Email Address: Forward your project materials to **projects@thinkcybergroup.com** and to your trainer.

Email Subject: Format your email subject as follows: **Project: PHISHING <Student Name:Unit Name>**. Make sure to replace **<Student Name>** with your actual name and **<Unit Name>** with the precise name of the unit as shown in the simulator.

4. **Unique Work**

   Your work **must** be unique. If the system detects that your project has been copied from another student, **both** the student who copied the work **and** the student who allowed their work to be copied will be **banned**. After being banned, the server will no longer check your future submissions.

   Important: You are encouraged to help each other with ideas and techniques, but each student **must submit their own, original project**. If you submit a project that is flagged as identical or too similar to another student's, both of you will be banned from submitting future work to the system.

5. **Use of AI Tools**

   You are allowed to use tools like ChatGPT or other AI bots for assistance, but your program must work and be clear. If your AI code is simply copied, doesn't work, or is unclear, you will be banned from future submissions.

Response Time: After submitting, you can track the status of your submission (pass or fail) through the ***Progress Tracker*** inside your student account. Allow up to one week for your project to be reviewed.

How to Showcase Your Cybersecurity Projects on LinkedIn

1. **Write a Compelling Post**:
   When you share your project on LinkedIn, make sure to write a post that highlights the importance of the project and your key takeaways from it. Use engaging language and explain:
   - **The challenge** you solved.
   - **Tools** and technologies you used (mention relevant tools like Python, Shell Scripting, Scapy, etc.).
   - **Skills** you developed (problem-solving, incident response, penetration testing, etc.).
   - **Real-world relevance** of your project. Explain how it can be applied to secure networks, detect vulnerabilities, or analyze threats in a professional setting.

Example:

"Excited to share my latest project in Network Security (ZX305) where I built a script to detect network anomalies. This project helped me understand intrusion detection and how to safeguard systems against real-world threats. Using Python and Shell scripting, I developed a tool that can assist security teams in monitoring and reacting to suspicious network activities."

2. **Attach Your PDF**:
   Upload the PDF file of your project as an attachment to your post. This will allow recruiters, potential employers, or other professionals in the cybersecurity field to directly review your work.
   - Ensure your **PDF is professional**, free of errors, and clearly explains each section of the project.
   - **Highlight results**: If you were able to detect or simulate cyber threats, mention how your tool worked and the outcomes of your project.

3. **Record a Video Demo**:
   Videos are powerful on LinkedIn. Record a quick demo of your project where you:
   - Explain the problem.
   - Walk through your code.
   - Show your script in action (for example, showing how your network security tool detects a threat).

Example Intro:

"Here's a quick demo of my project where I developed a Python-based tool to monitor network traffic and detect potential cyber threats. Watch how it scans real-time traffic and flags suspicious activity."

4. **Mention the Certificate**:
   Once you complete the project, your simulator generates an official **ThinkCyber certificate**. Share this alongside your post to demonstrate your verified completion of the project.

Example:

"Proud to have completed my Network Security (ZX305) program with ThinkCyber! This certificate is proof of the skills I've acquired in identifying and mitigating cybersecurity threats."



Make sure to upload the certificate as an image. Use relevant hashtags like **#cybersecurity**, **#networksecurity**, and **#Python** to increase your visibility.

5. **Encourage Networking**:
   Mention that you are open to connecting with cybersecurity professionals and recruiters who might find your skills and projects relevant to their needs.